

for transfer to a requestor.

In order for different vendors to participate as widely as possible in the distribution of Software Images for the widest variety of products without requiring bilateral distribution agreements between each pair, it is RECOMMENDED for vendors to participate in distribution schemes that maximize availability across other vendors and OTA Providers.

Vendors SHOULD build data sets aggregating the metadata and payloads of Software Images to support their OTA Image Selection Logic by any means they deem satisfactory. Vendors SHOULD refer to canonical databases, such as the [Distributed Compliance Ledger](#).

Given that most Fabrics likely will contain a reduced subset of Nodes capable of acting as OTA Providers compared to the larger set of vendors represented in the many deployed Nodes, it is advantageous to end-users that Vendors attempt to cover as many other vendors with their data sets. This will ensure that the majority of Software Image queries can be fulfilled if a vendor has released a newer version than that installed on the querying OTA Requestor.

### Download Protocol selection

The OTA Image Selection Logic SHALL consider the OTA Requestor's [supported download protocols](#) to determine whether to respond to a [QueryImage](#) command.

If either the BDXSynchronous or BDXAsynchronous protocols are supported by the OTA Requestor, the OTA Provider SHALL prefer to respond to the OTA Requestor with a BDX protocol URI, as long as it can fulfill the role of BDX server for the OTA Requestor.

Otherwise, if the HTTPS protocol is supported by the OTA Requestor, and the OTA Provider determines that an OTA Software Image is available to fulfill the request from a server supporting HTTPS, it SHOULD respond with the direct source URI, so that the OTA Requestor MAY download it directly.

Otherwise, if the VendorSpecific protocol is supported by the OTA Requestor, and the OTA Provider has sufficient knowledge of the OTA Requestor's capabilities based on the [QueryImage](#) command arguments, it SHOULD respond with a URI which is known to be understood by the OTA Requestor. It is RECOMMENDED to limit usage of this modality and prefer BDXSynchronous and BDXAsynchronous.

### Conceptual algorithm for matching OTA Software Images applicable to a query

An OTA Provider MAY use any of the fields of the [QueryImage](#) command in any way it deems applicable to determine whether an appropriate OTA Software image is available for the OTA Requestor.

However, to increase interoperability, OTA Providers which have access to the data present in the [Distributed Compliance Ledger \(DCL\)](#), whether from cached subset or from a full replica, SHOULD employ at least the common conceptual algorithm provided in this section to determine whether an OTA Image is available.

The information to access OTA Software Image locations for certified software versions is available in the [DCL DeviceSoftwareVersionModel Schema](#), which is indexed by VendorID, ProductID and